Introduction

R0005E - Measurement and Feedback Control

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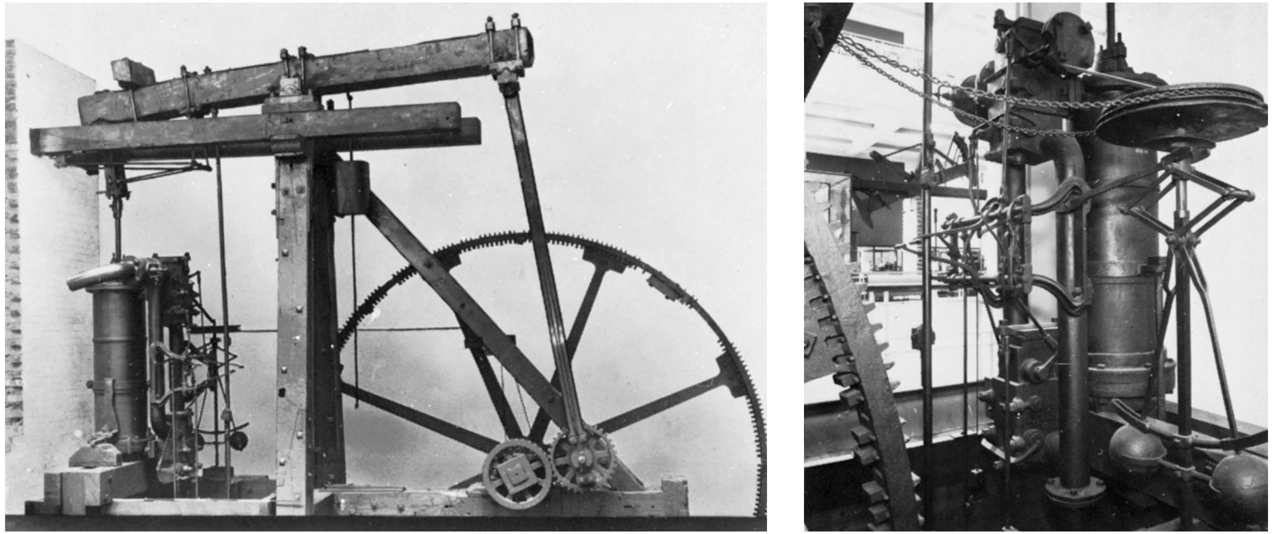
Historic example

Steam engine

- Around 1788 James Watts designed and applied a fly-ball governor to a steam engine.
- Thereby, the speed of the engine could be controlled.



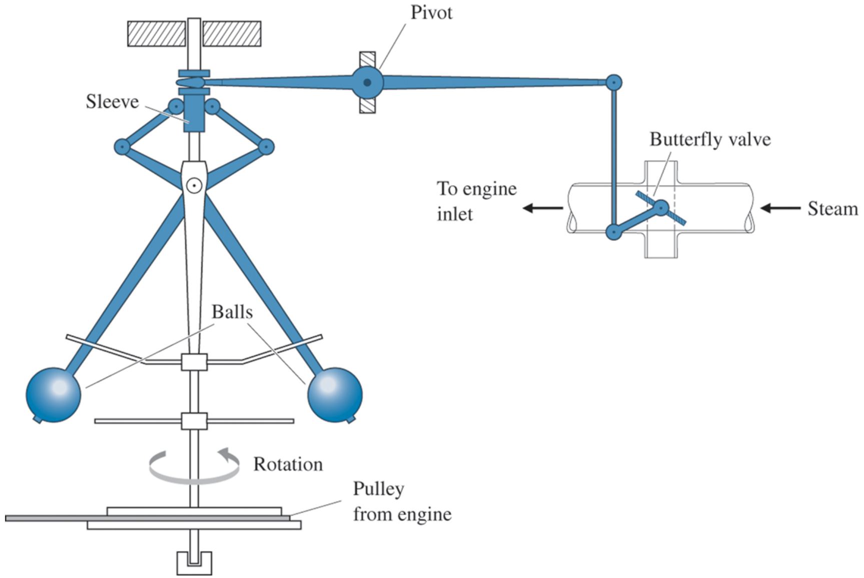
Source: Science Museum, London.

But:

- How does it work?
- What is the underlying principle of its function?

Fly-ball governor

The governor is a mechanical controller



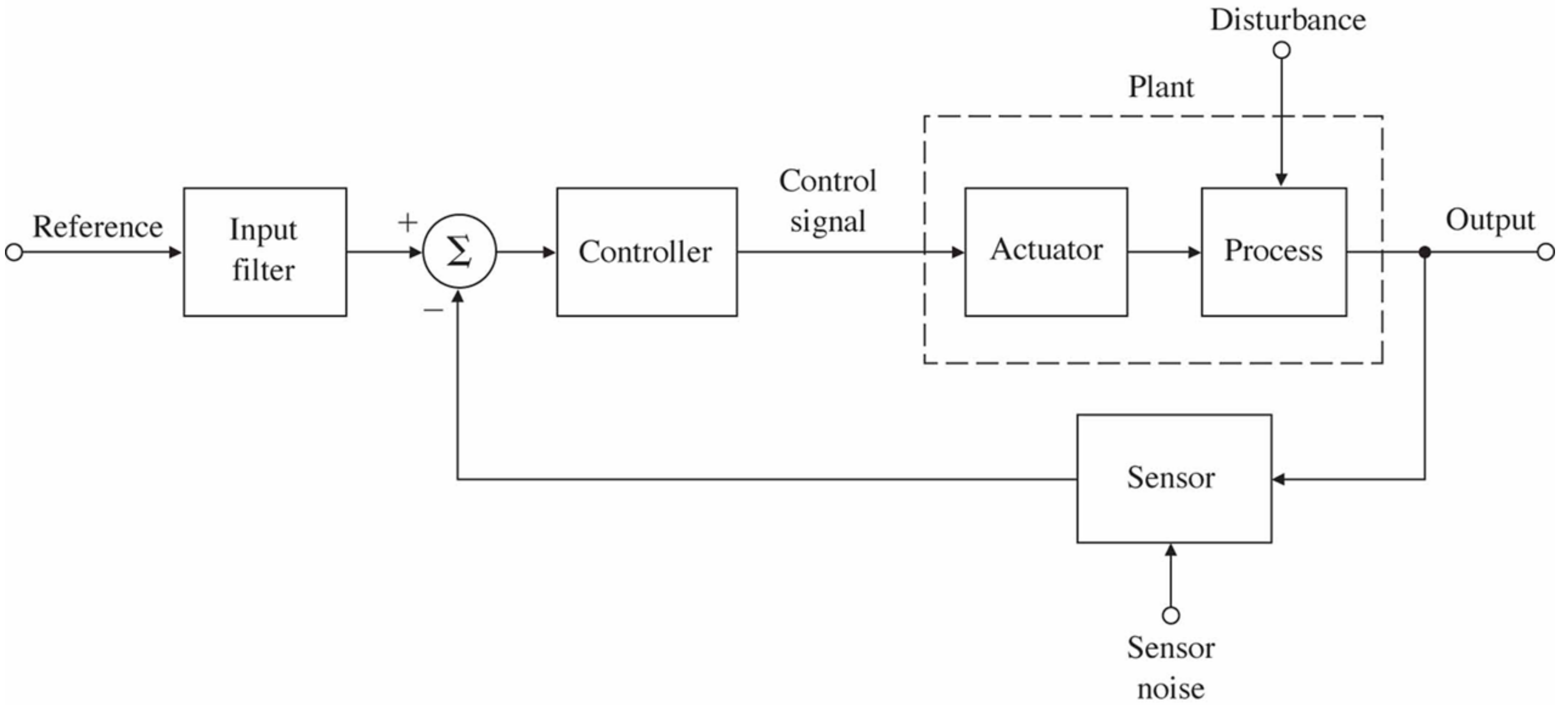
Source: Feedback Control of Dynamic Systems, Franklin et. al.

Components:

- Sensor: Pulley together with the fly-balls
- Comparator and set point adjustment: Sleeve
- Controller: Pivot
- Actuator: Butterfly valve

Typical components of a control system

Now we can compose a block diagram of the control system:

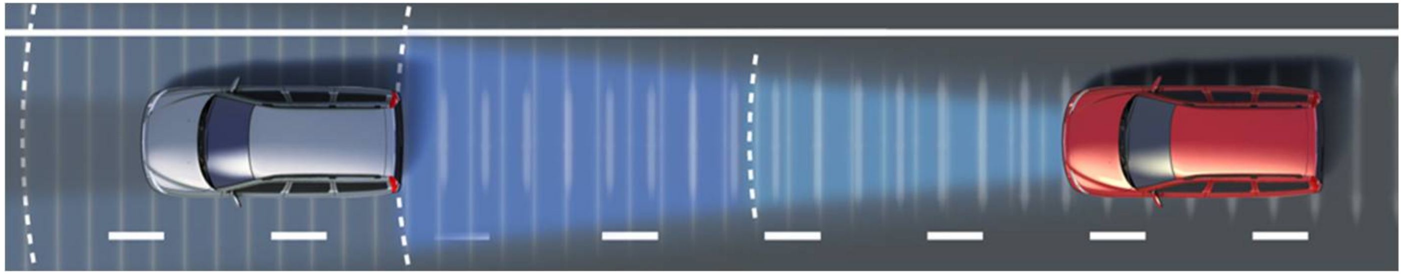


Source: Feedback Control of Dynamic Systems, Franklin et. al.

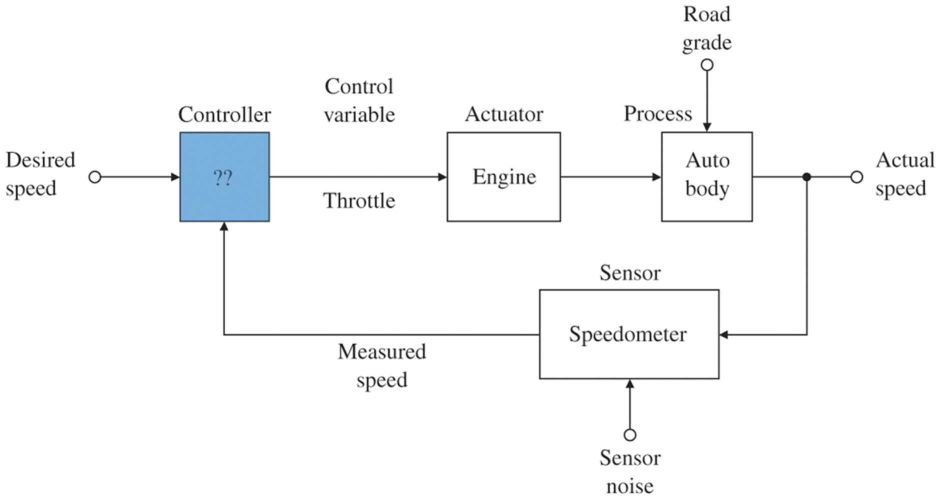
- These block can be found in any control system
- Systematic way to describe control systems
- And: it is generally applicable

Analysis of a feedback system: Cruise Control

A cruise control is a standard feature of a modern car. Let's have a closer look how it works.



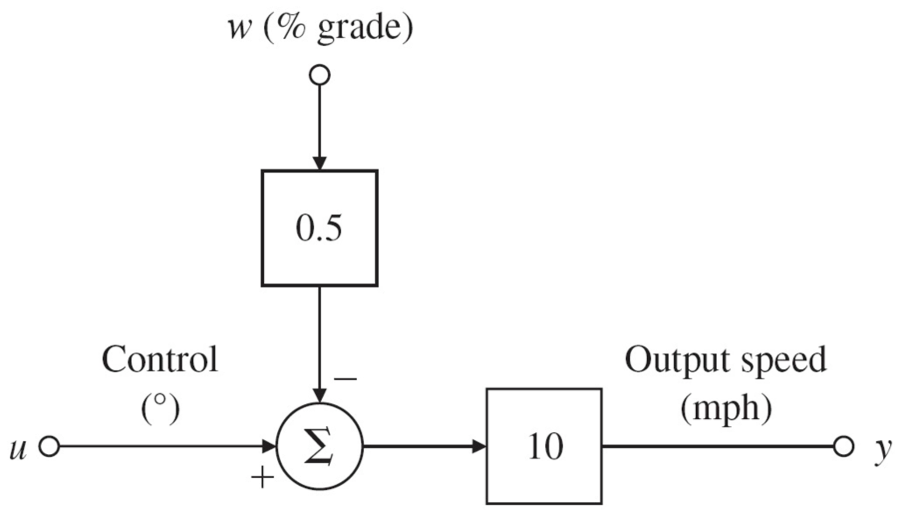
We can represent it in a general block diagram:



Source: Feedback Control of Dynamic Systems, Franklin et. al.

Analysis of a feedback system: Cruise Control (2)

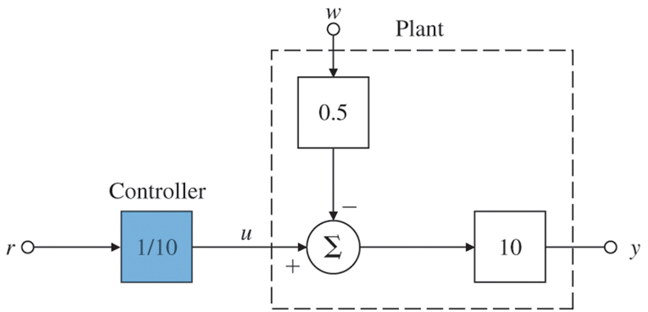
We analyze how the car is reacting on changes in the throttle angle and get a model for the car. Additionally, we analyze how the road grade is affecting. This can be combined into the following model of the plant:

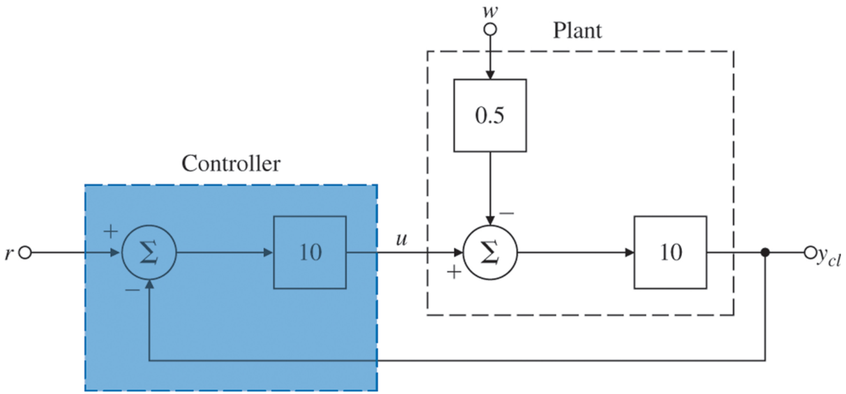


There are now two ways to control:

- Open loop: Determine the throttle angle for a desired speed based on the model.
- Closed loop: Make use of a speed measurement and adjust the throttle angle.

Block diagram of the two concepts:





Source: Feedback Control of Dynamic Systems, Franklin et. al.

Comparison:

- How well does the car follow the desired reference speed?
- How does the road grade affect the speed of the car?
- What are the pros and cons?

Conclusion:
The control system could be analyzed by assessment of the signals and their dependencies!

Open loop control:

- Very good reference tracking
- Very bad disturbance rejection

Closed loop control:

- Good reference tracking
- Good disturbance rejection

Some questions:

- How can I design the controller?
- Is it possible to know how the system will behave?
- How can I design a controller that makes the system efficient?

Note: We have only considered the static behavior!

Additional material

Brian Douglas has a very nice channel on YouTube[SE] which provides complementary information to the lectures. The videos are very helpful provide a potentially different view!

His channel is https://www.youtube.com/user/ControlLectures. I will point to some of his videos during the course.

- Why Learn Control Theory?
  https://www.youtube.com/watch?v=oBc_BHxw78s
- Control Systems Lectures - Closed Loop Control
  https://www.youtube.com/watch?v=O-OqgFE9SD4